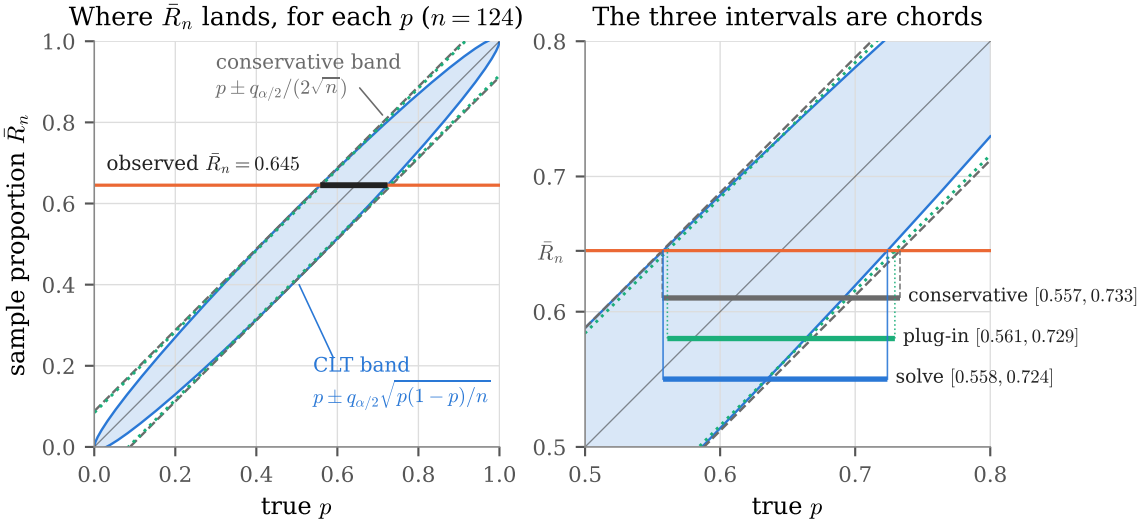

Where $\bar{R}_n$ lands, for each p ($n = 124$)
conservative band
$p \pm q_{\alpha/2}/(2\sqrt{n})$
observed $\bar{R}_n = 0.645$
CLT band
$p \pm q_{\alpha/2}\sqrt{p(1-p)/n}$
sample proportion $\bar{R}_n$
true p
The three intervals are chords
$\bar{R}_n$
conservative $[0.557, 0.733]$
plug-in $[0.561, 0.729]$
solve $[0.558, 0.724]$
true p